ATOM COUNT CARD

MOLE CONCEPT ACTIVITY 2

Baking Soda ($NaHCO_3$): From a printed label to moles, formula units, and atoms

Quantity



A PRODUCT IDENTITY

Product: Baking Soda (100% Sodium Bicarbonate, 250 g net weight)

Compound: Sodium bicarbonate ($NaHCO_3$)

Why it matters: An essential leavening agent in food chemical reactions and a versatile household buffer/cleaning compound

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B BASE QUANTITATIVE DATA

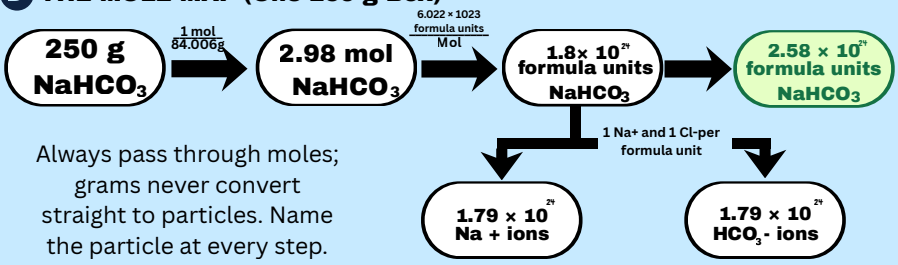
Quantity	Value	Source / Note	Precision
Net content ($NaHCO_3$)	250g	Product label	2s.f. (Limits precision)
Molar mass of $NaHCO_3$	84.006g/mol	CIAAW / PubChem (CID 516892)	Not limiting
Avogadro constant (N_A)	$6.02214076 \times 10^{23} mol^{-1}$	BIPM (2019)	Exact
Atomic mass Na, H, C, O	22.990, 1.008, 12.011, 15.999 g/mol	CIAAW standard abridged table	Not limiting

Assumptions: Pure sodium bicarbonate (100% NaHCO), no added anti-caking agents.

C MOLAR MASS BUILD-UP

Element	Atoms	g/mol each	Total Contribution (g/mol)	% by Mass
Na	1	22.990	22.990	27.367%
H	1	1.008	1.008	1.200%
C	1	12.011	12.011	14.298%
O	3	15.999	47.997	57.135%
Total	6	____	84.006	100.000%

D THE MOLE MAP (One 250 g Box)



Always pass through moles; grams never convert straight to particles. Name the particle at every step.

E MOLE-CONVERSION CALCULATIONS

1. Molar Mass Build-up:

$$M = 22.990 + 1.008 + 12.011 + 3(15.999) = 84.006 \text{ g/mol } NaHCO_3$$

2. Label Bridge (Printed amount to pure mass):

$$250 \text{ g package} \times \frac{100 \text{ g } NaHCO_3}{100 \text{ g product}} = 2.5 \times 10^2 \text{ g } NaHCO_3 \quad (2 \text{ s.f.})$$

3. Mass → Moles:

$$250 \text{ g } NaHCO_3 \times \frac{1 \text{ mol } NaHCO_3}{84.006 \text{ g } NaHCO_3} = 2.9759 \text{ mol} \rightarrow 3.0 \text{ mol } NaHCO_3$$

4. Moles -> Formula Units:

$$2.9759 \text{ mol} \times \frac{6.02214076 \times 10^{23} \text{ formula units}}{1 \text{ mol}} = 1.7921 \times 10^{24} \rightarrow 1.8 \times 10^{24} \text{ formula units } NaHCO_3$$

5. Atoms of One Element (Oxygen inside the formula):

$$1.7921 \times 10^{24} \text{ formula units} \times \frac{3 \text{ O atoms}}{1 \text{ formula unit}} = 5.3763 \times 10^{24} \rightarrow 5.4 \times 10^{24} \text{ O atoms}$$

6. Percent Composition & Two-Route Check (Sodium, Na):

$$\%Na = \frac{22.990 \text{ g/mol}}{84.006 \text{ g/mol}} \times 100\% = 27.367\%$$

Route A (Mass x %):

$$250 \text{ g} \times 0.27367 = 68.42 \text{ g Na}$$

Route B (Moles x Ratio x Molar Mass):

$$2.9759 \text{ mol } NaHCO_3 \times \frac{1 \text{ mol Na}}{1 \text{ mol } NaHCO_3} \times 12.011 \text{ g/mol} \times \frac{22.990}{12.011} = 68.42 \text{ g Na}$$

Both routes agree (68g Na at 2 sy.f.).

7. Reverse: Percent Composition ->Empirical Formula:

In 100g: $Na: \frac{27.367 \text{ g}}{22.990} = 1.1904 \text{ mol}$ $H: \frac{1.200 \text{ g}}{1.008} = 1.1905 \text{ mol}$

$C: \frac{14.298 \text{ g}}{12.011} = 1.1904 \text{ mol}$ $O: \frac{57.135 \text{ g}}{15.999} = 3.5712 \text{ mol}$

Divide by smallest (1.1904):

$$Na = 1.000, H = 1.000, C = 1.000, O = 3.000 \rightarrow NaHCO_3$$

(Empirical = Molecular, n=1)

8. Scale-up & Reverse:

Scale-up (20 boxes for a lab):

$$20 \times 2.9759 \text{ mol} = 59.518 \text{ mol} \rightarrow 6.0 \times 10^1 \text{ mol } NaHCO_3$$

Reverse (Packages for 1.00\text{ mol}):

$$1.00 \text{ mol} \times \frac{84.006 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ package}}{250 \text{ g}} = 0.3360 \rightarrow 0.34 \text{ package}$$

F SCALE-UP: FROM ONE PACKAGE TO THE WHOLE CAMPUS

ONE PACKAGE (250 g):

$3.0 \text{ mol} \mid 1.8 \times 10^{24}$ formula units

ONE LAB (20 packages / 5 kg):

$6.0 \times 10^1 \text{ mol} \mid 3.6 \times 10^{25}$ formula units

CAMPUS (100 packages / 25 kg):

$3.0 \times 10^2 \text{ mol} \mid 1.8 \times 10^{26}$ formula units

REVERSE: 1.00 mol of $NaHCO_3$ is

contained in 0.34 package (approx. 1/3 of the box).

G SIGNIFICANT-FIGURE CHECK & ONE-NUMBER TAKEAWAY

Sigfig Check: 250 g sets the limit to 2 significant figures (2.5×10^2 g). Avogadro's constant and formula subscripts are exact and non-limiting.

One-Number Takeaway: One 250 g box of baking soda contains about 1.8×10^{24} formula units, yet a single mole (6.022×10^{23} units) weighs only 84.006 g

H REFERENCES & AI DISCLOSURE

PubChem. (n.d.). Sodium Bicarbonate (CID 516892). National Center for Biotechnology Information. https://pubchem.ncbi.nlm.nih.gov/compound/Sodium-Bicarbonate

CIAAW. (2024). Abridged standard atomic weights. IUPAC. https://www.ciaaw.org/abridged-atomic-weights.htm

AI Use Statement: AI was used to assist in organizing the data layout and reviewing the significant-figure rounding for this card. All calculations, chemical formulas, and values were independently verified.

AI Use Disclosure

I used AI (Gemini) to assist with brainstorming layout ideas, structuring the infographic format, and polishing the text formatting for this Atom Count Card. All label data, atomic masses, dimensional analysis, step-by-step mole conversions, significant figure decisions, and final calculations were independently performed and verified by the student.

References

1. Bureau International des Poids et Mesures. (2019). *The International System of Units (SI)* (9th ed.). https://www.bipm.org/en/publications/si-brochure

2. Commission on Isotopic Abundances and Atomic Weights. (2024). *Abridged standard atomic weights*. International Union of Pure and Applied Chemistry. https://www.ciaaw.org/abridged-atomic-weights.htm

3. National Center for Biotechnology Information. (n.d.). *PubChem compound summary for CID 516892, Sodium bicarbonate*. Retrieved October 13, 2026, from https://pubchem.ncbi.nlm.nih.gov/compound/Sodium-Bicarbonate